

Halstead Metrics Practice Problems

Question 1:

```
#include <stdio.h>

// Program to add two numbers
int main() {
    int a = 10, b = 20, sum;
    sum = a + b;
    printf("Sum is: %d\n", sum);
    return 0;
}
```

Identify the operators and operands in the program above. **Calculate** the following Halstead metrics: n_1 , n_2 , N_1 , N_2 , Vocabulary, Length, Program Volume, Effort and Difficulty.

Question 2:

```
#include <stdio.h>

// Program to find the factorial of a number
int main() {
    int i, fact = 1, n = 5;
    for(i = 1; i <= n; i++) {
        fact = fact * i;
    }
    printf("Factorial: %d\n", fact);
    return 0;
}
```

Identify the operators and operands in the program above. **Calculate** the following Halstead metrics: n_1 , n_2 , N_1 , N_2 , Vocabulary, Length, Program Volume, Effort and Difficulty.

Question 3:

```
#include <stdio.h>
```

```
// Program to find the largest of two numbers
```

```
int main() {  
    int x = 15, y = 25, max;  
    if(x > y) {  
        max = x;  
    } else {  
        max = y;  
    }  
    printf("Maximum: %d\n", max);  
    return 0;  
}
```

Identify the operators and operands in the program above. **Calculate** the following Halstead metrics: n_1 , n_2 , N_1 , N_2 , Vocabulary, Length, Program Volume, Effort and Difficulty.

Question 4:

```
#include <stdio.h>
```

```
// Program to calculate the sum of even numbers from 1 to 10
```

```
int main() {  
    int i, sum = 0;  
    for(i = 1; i <= 10; i++) {  
        if(i % 2 == 0) {  
            sum = sum + i;  
        }  
    }  
    printf("Sum of even numbers: %d\n", sum);  
    return 0;  
}
```

Identify the operators and operands in the program above. **Calculate** the following Halstead metrics: n_1 , n_2 , N_1 , N_2 , Vocabulary, Length, Program Volume, Effort and Difficulty.